SHADAB KHEDRI
SUPERVISOR: YANYAN SHA

Effect of risk reducing measures to re the design loads for ship impact for tl floating bridge

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Faculty of Science and Technology

Department of Mechanical, Construction and Materials Technology

Abstract

This thesis presents a probabilistic assessment of ship-bridge collision risk for a proposed floating suspension bridge across the Nordfjord in western Norway. The study focuses on estimating the likelihood and severity of ship allisions with the bridge tower using one year of Automatic Identification System (AIS) data collected between October 2023 and 2024. The bridge configuration includes a single central support tower located mid-span, dividing the navigational channel into two opposing traffic lanes.

The ship traffic was filtered, classified by direction (eastbound and westbound), and projected along the bridge axis. Kernel Density Estimation (KDE) was used to model the lateral distribution of crossing positions (f(z)), which allowed for a data-driven, non-parametric representation of vessel trajectories. This KDE was then applied within Pedersen's four-category collision framework, focusing on: (1) ordinary route deviations due to human error, (2) missed course corrections at turning points, (3) technical failures and drifting events, and (4) evasive maneuvers (Category 4 was excluded due to enforced traffic separation).

The analysis was performed separately for traffic in both directions. However, due to the fjord's geometry and dead-end configuration, the inbound and outbound vessel distributions were found to be nearly symmetrical, allowing westbound results to represent both directions. The risk-reducing effects of Vessel Traffic Services (VTS) and Traffic Separation Schemes (TSS) were also modeled using fixed reduction factors applied to key probability parameters.

Results show that Category 2 (failed turns) dominates the total risk. The total unmitigated collision probability over a 100-year design life was estimated at 4.14%. With VTS/TSS measures in place, this was reduced to 2.07%. The design ship, selected based on the vessel class with the highest risk-weighted impact contribution, was a 340 m LOA ship. Impact loads were evaluated using an energy-based model that incorporated real AIS-derived crossing velocities. The unmitigated design force was 405 MN, while the expected impact load under mitigation dropped to 8.4 MN—supporting more cost-efficient and reliable bridge design strategies.

This work demonstrates how integrating real-world AIS data, probabilistic modeling, and mitigation effects can support performance-based structural design of floating bridges in constrained maritime corridors.

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List of Symbols and Abbreviations

Abbreviation	Description
AIS	Automatic Identification System
KDE	Kernel Density Estimation
LOA	Length Overall
COG	Course Over Ground
VTS	Vessel Traffic Services
TSS	Traffic Separation Scheme
PDF	Probability Density Function
IWRAP	International Waterway Risk Assessment Program
DNV	Det Norske Veritas
AASHTO	American Association of State Highway and Transportation Officials
CEN	European Committee for Standardization (Comité Européen de Normalisation)
MN	Meganewton (1 MN = 10^6 N)
MJ	Megajoule (1 MJ = 10^6 J)

Symbols

Symbol	Description
$f(z)$	Lateral probability density function of ship crossings
$Q_i(t)$	Number of ship passages in class ii at time t
P_c	Causation factor (probability of collision given collision course)
P_t	Probability of totally neglecting to change course
P_d	Probability of failing to verify ship position
P_0	Probability of omission (Category 2)
d	Distance traveled on incorrect course
a	Navigator attention interval
$\mathrm{P_e}$	Probability of failed evasive action or drifting event
T	Time period or design life (typically 1 year)
$\mathrm{P_{Cat.1}}$	Probability of Category 1 collision
$\mathrm{P_{Cat.2}}$	Probability of Category 2 collision
$\mathrm{P_{Cat.3}}$	Probability of Category 3 collision
$\mathrm{E_k}$	Kinetic energy of the ship at impact
m	Mass of the ship
$\mathrm{k_a}$	Added mass coefficient (accounts for surrounding water mass)
m_{eff}	Effective mass including added mass
$\mathrm{V_n}$	Ship velocity normal to the bridge at impact
F_{impact}	Peak dynamic impact force
R	Structural resistance of the bridge pier
s	Domain scaling factor for KDE rescaling
z	Lateral position across the navigation corridor

1. Chapter1: Introduction

1.1 Background

As Norway expands its transport infrastructure across its many deep fjords, floating bridges have become a promising solution where conventional fixed bridges are not feasible. These structures, however, introduce new safety considerations—particularly the risk of ship-bridge collisions in navigable waterways. This concern is especially important for suspension bridges with mid-span towers, where bridge piers lie directly within active ship lanes.

Traditional design codes, such as Eurocode EN 1991-1-7 and AASHTO, provide guidance for accounting for accidental ship impact loads. However, these guidelines were originally developed for inland, fixed-pier bridges and may not capture the complexity of dynamic marine environments or long-span floating systems. As such, there is a growing need for data-driven, probabilistic approaches that can incorporate local marine traffic behavior into bridge design.

In this context, Automatic Identification System (AIS) data offers a valuable source of empirical information for modeling vessel movement. By analyzing real-world ship trajectories, designers can estimate the probability of collision and the associated impact loads more accurately, enabling more robust and risk-informed structural design.

1.2 Problem Statement

The planned floating suspension bridge across Nordfjord introduces a mid-span pier into an active shipping lane. While this configuration may be structurally advantageous, it increases the likelihood of allision events from passing ships. Without accurate, site-specific modeling of collision probabilities and impact forces, the bridge design may either be under-conservative (leading to safety risks) or overly conservative (leading to unnecessary costs).

Moreover, existing standards do not explicitly account for symmetrical traffic flow in dead-end fjords**,** or the mitigating effects of Vessel Traffic Services (VTS) and Traffic Separation Schemes (TSS). These factors must be incorporated into a refined probabilistic framework that considers actual vessel movements, ship class distributions, and heading behavior near the bridge.

1.3 Research Objectives

The primary objective of this thesis is to perform a probabilistic risk analysis of ship collisions with a proposed floating bridge across Nordfjord, with the following specific goals:

- To classify and quantify ship collision risks using Pedersen's four-category model
- To process and analyze AIS data to determine ship crossing distributions and directional flow
- To evaluate the impact of VTS and TSS on collision probability
- To determine the design ship and estimate impact loads under mitigated and unmitigated conditions
- To inform risk-based design decisions for bridge towers in fjord environments

1.4 Significance of the study

This study contributes to both the theoretical and practical understanding of ship-bridge collision risk in offshore bridge design. From a practical standpoint, it offers a method to incorporate real traffic data and operational controls into the design process. From a research perspective, it demonstrates the application of a probabilistic collision framework to a marine context with symmetrical bidirectional traffic—a case not directly covered by existing design standards.

Furthermore, the findings can support future guidelines for floating bridge projects in Norway and other fjord regions, where safe integration with marine traffic is a critical design challenge.

1.5 Assumptions and Limitations

To perform this study within the available data and time constraints, the following assumptions and limitations were adopted:

- Human error and drifting failure probabilities were modeled using values from literature (e.g., Pedersen, IWRAP Mk II) rather than extracted from local incident reports.
- No significant heading changes were detected in AIS data near the bridge, likely due to low maneuvering frequency or sampling rate limitations. Thus, Category 2 was modeled using assumed parameters.
- Category 4 collisions (evasive maneuver interactions) were excluded, based on the implementation of a traffic separation scheme that ensures eastbound and westbound vessels use opposite lanes.
- Eastbound and westbound risks were treated as equivalent due to the dead-end nature of Nordfjord, where all vessels entering must later exit.

These assumptions introduce some modeling uncertainty, but they reflect a conservative and realistic approach grounded in existing literature and site-specific constraints.

1.6 Structure of the Report

This thesis is organized into six chapters:

- Chapter 1 introduces the background, objectives, and scope of the study.
- Chapter 2 reviews relevant literature on ship domain modeling, maritime risk frameworks, bridge design codes, and impact load estimation.
- Chapter 3 presents the methodology, including AIS data processing, probabilistic modeling, and KDE-based lateral distribution analysis.
- Chapter 4 summarizes the probabilistic results for each collision category and compares the effects of VTS/TSS.
- Chapter 5 identifies the design ship and computes impact loads under both mitigated and unmitigated conditions.
- Chapter 6 concludes the study and offers recommendations for future research and design practices.

2 Chapter2: Literature Review

2.1 Methods Using the Ship Domain Concept

The concept of the *ship domain* remains central to collision risk assessment in maritime contexts. Initially introduced by Fujii and Tanaka (1971) as the "effective domain" and later refined by Goodwin (1975), the ship domain represents the spatial buffer that navigators instinctively maintain around their vessels to avoid collisions.

Subsequent research has emphasized the domain's dependency on operational context. Studies by Gucma and Marcjan (2012), Hansen et al. (2013), and Szlapczynski (2017) demonstrate how ship size, waterway geometry, traffic density, and encounter types influence the effective domain. Their findings reinforce the need for localized domain modeling—particularly in complex waterways like Norwegian fjords.

Recent advances in Automatic Identification System (AIS) technology have enabled the extraction of dynamic ship domains based on real navigational behavior. For example, Zhang et al. (2016, 2017) proposed probabilistic models such as the VCRO and MDTC, which incorporate ship behavior, maneuverability, and intersection angles to evaluate collision proximity. These tools have gained traction in both human-operated and autonomous navigation systems (Du et al., 2020; Huang et al.,

2020). Nevertheless, concerns remain regarding their predictive accuracy, as noted by Montewka et al. (2020) and Rawson and Brito (2020), who caution that many domain models reflect post-avoidance trajectories rather than true collision risk.

In the context of this thesis—which evaluates ship allision risk for a floating suspension bridge with a mid-span pier—direction-specific domain modeling is crucial. Given the bidirectional traffic pattern (eastbound and westbound), the application of AIS-informed ship domain analysis allows for more precise risk estimation and targeted design interventions.

2.2 Simulation Models Used in Maritime Risk Assessment

Simulation has long played a pivotal role in maritime risk assessment, evolving from early offshore engineering applications to sophisticated, real-time traffic simulators. Källström and Ramzan's (1985) pioneering work on TLP installation laid the groundwork for using simulation in complex marine environments.

Today's simulation tools serve three major purposes:

1. **Macro-level risk estimation**, often through models like MTS (Maritime Transportation System) and its derivative DMTS
2. **Autonomous navigation simulation**, incorporating reactive maneuvering under environmental constraints
3. **Hydrodynamic and operational planning**, used in port logistics and offshore construction

MTS and DMTS models rely on AIS trajectory data and probabilistic injection of failure modes—derived from both historical records and expert judgment (Ulusçu et al., 2009; Goerlandt & Kujala, 2011). These models are valued for their efficiency and flexibility in evaluating large-scale traffic scenarios, including human error and environmental factors. The ShipRisk software (Rasmussen et al., 2012), which integrates Pedersen's causation logic with MTS principles, has been widely adopted for infrastructure projects such as the Fehmarnbelt link and is adaptable to floating bridge scenarios.

Other simulators, such as Shen et al. (2019), focus on autonomous path planning under hydrodynamic constraints. Full-mission bridge simulators remain essential for training and local risk evaluations (Chen et al., 2018; Baldauf & Benedict, 2018), providing human-in-the-loop feedback for decision-making scenarios.

In this thesis, MTS-style simulations are employed to model realistic traffic flows near the bridge pier, incorporating AIS-informed trajectories and probability-based failure logic. Though not hydrodynamically detailed, these models are well-suited for estimating design-level impact probabilities during early-stage planning.

2.3 Bridge Design Codes and Probability of Allision

Structural design standards such as Eurocode EN 1991-1-7 (CEN, 2006) and the AASHTO Guide Specification for Vessel Collision Design of Highway Bridges (AASHTO, 2009) offer foundational frameworks for assessing accidental loads on bridges due to ship impact. These frameworks adopt probabilistic formulations, accounting for the frequency of vessel transits, likelihood of collision (causation factor), and the mitigating effect of evasive actions or navigational control systems (e.g., VTS). AASHTO and EUROCODE approaches explained in section 2.4.1.

While these models are well-established, they were primarily formulated for fixed bridges in inland waterways, such as river crossings. Applying them directly to floating bridge structures in dynamic marine environments (e.g., fjords with tidal currents, wind, and long-span oscillation) presents limitations. These include the lack of lateral confinement, variable ship approach angles, and bridge flexibility—all of which affect the collision dynamics and failure modes.

As a result, more advanced probabilistic methods—such as Pedersen's category-based model and modern AIS-informed simulation frameworks (e.g., IWRAP MkII)—are increasingly used to adapt risk calculations to complex geographies. These tools allow the integration of real traffic patterns, lateral ship distributions, and failure probabilities, resulting in more accurate and context-sensitive estimation of ship impact loads.

Pedersen's Four Collision Categories: Expanded Definitions

according to Pedersen's model, Ships can hit the bridge line due to certain erroneous ship track patterns. To systemize the different possibilities for ships to hit the bridge line, the model to estimate the collision frequency will here be based on the sum of four accident categories:

$$P_{total} = P_{Cat.1} + P_{Cat.2} + P_{Cat.3} + P_{Cat.4} \qquad \textit{(Equation 2-1)}$$

The collision frequency estimation can be based on an evaluation of four accident categories:

Accident Categories:

Category 1 – Ordinary Route Collisions

Category 1 collisions represent events in which vessels traveling along their planned navigational route deviate laterally—often due to human error—and strike a bridge

structure. These deviations may result from lapses in attention, misjudgment of position, or inadequate response to environmental conditions.

The likelihood of such events is typically modeled using the lateral probability distribution f(z), which quantifies the spatial spread of ship crossings along a bridge axis. This distribution is commonly derived from historical AIS data and fitted to probability density functions such as Gaussian, Lognormal, or Weibull (Pedersen, 1995; Chen et al., 2019). A small fraction of the total traffic is assumed to represent gross navigation errors—unusual but high-consequence deviations from the main track—typically estimated at 1–5% of observed crossings (Montewka et al., 2010).

$$P_{Cat.1} = \frac{1}{T}\sum_{ship\ class\ i}^{n} \int_{Z=Z_a}^{Z_b} \int_0^T P_{c,i}\, Q_i(t)\ f(z)\, dt\, dz \qquad \textit{(Equation 2-2)}$$

Where:

- i =the ship class (for instance ships with lengths between 130 and 150 m).
- Z= an axis transverse to the shipping route. Za , Zb denotes integration limits, see Figure 2.1
- $P_{c,i}$ = the causation factor, i.e. the probability of a collision if the ship is on collision course.
- $Q_i(t)$= the number of ships of class i per time unit
- $f(z)$ = the lateral ship distribution.

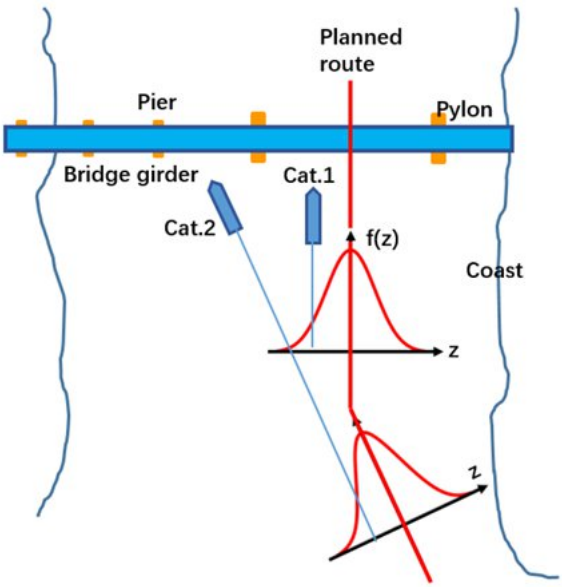


Figure 2-1 Category 2 – Failed Course Changes at Turning Points

Category 2 collisions occur when a ship fails to perform a required turn at or near a navigational bend, continuing on an incorrect heading toward the bridge. This type of failure is particularly critical in areas where the sailing route shifts laterally to avoid a central pier or support.

$$P_{Cat.2} = \frac{1}{T}\sum_{ship\ class\ i}^{n} \int_{A}^{B} \int_{0}^{T} P_0\, P_{c,i}\, Q_i(t)\ f(z)\, dt\, dz \qquad \textit{(Equation 2-3)}$$

The probability of such omission is defined by Pedersen's formula:

$$P_0 = P_t + (1 - P_t) \times P_d^{\frac{d-a}{a}} \qquad \textit{(Equation 2-4)}$$

Where:

- P_t = probability of completely neglecting to initiate a turn
- P_d = probability of failing to verify the ship's position
- d = distance traveled on the incorrect course
- a = average interval between navigational checks

This equation reflects both intentional neglect and unintentional position-check failure. In practical applications, parameters are assigned based on prior studies or calibrated using observed ship behavior through AIS data (Goerlandt & Kujala, 2011; Svanberg et al., 2019). In this thesis, no significant heading changes were observed near the bridge, so assumed constants were used: $P_t = 0.01$, $P_d = 0.1$, $d = 500m$, and $a = 1000m$, reflecting conservative assumptions for confined waters.

Category 3 – Technical Failures and Drifting

Category 3 represents collisions caused by ship system failures (e.g., loss of propulsion or steering) or environmental drift. These are non-navigational failures and are typically rare but unpredictable. A drifting vessel may collide with the bridge if it cannot anchor effectively or if wind and current conditions push it toward the structure before recovery is possible.

$$P_{Cat.3} = \frac{1}{T}.P_{c,i}\ .P_e \sum_{ship\ class\ i}^{n} \int_{Z=Z_a}^{Z_b} \int_{0}^{T} Q_i(t)\ f(z)\, dt\, dz \qquad \textit{(Equation 2-5)}$$

The probability of allision in this category depends on:

- The initial failure probability
- The success of anchoring (often modeled with a fixed success rate, e.g., $P_{anchor} = 0.7$
- The duration of the drift, commonly modeled by a Weibull distribution of repair time (DNV, 2013)
- The spatial crossing density $f(z)$ along the bridge

Many studies, including those using IWRAP MkII and DNV guidelines, assume a base value of $P_t = 10^{-3}$ for the probability that a drifting vessel crosses the bridge axis without regaining control (Friis-Hansen & Simonsen, 2002). This is adjusted

downward when VTS/TSS or passive mitigation systems (e.g., physical barriers) are in place.

Category 4 – Evasive Maneuver Collisions

Category 4 accounts for reactive maneuvers—typically last-minute course alterations due to crossing traffic, emergency avoidance, or unexpected objects. These maneuvers can inadvertently redirect a vessel toward the bridge structure. Though difficult to model directly, this category is important in high-traffic zones or where eastbound and westbound lanes are not physically separated.

The risk is often modeled using the frequency of close encounters, extracted from AIS trajectory crossings or predicted using ship domain simulations (Zhang et al., 2017). When eastbound and westbound traffic are routed on separate sides of the bridge—as assumed in this thesis—the occurrence of Category 4 events can be considered negligible.

2.4 Impact Load Models and Risk-Based Design

Bridge structures exposed to marine traffic must be designed to withstand accidental impacts from ships. The estimation of “design impact loads” is a critical component in ensuring the safety and serviceability of such infrastructure. Traditionally, deterministic methods were used to define impact forces based on the largest foreseeable vessel. However, modern design philosophy—especially in high-traffic or constrained navigational zones—has shifted toward “probabilistic risk-based approaches”.

2.4.1 Eurocode and AASHTO Approaches to Ship collision Risk

Design codes such as Eurocode (CEN, 2006) and AASHTO (2009) introduce probability-based formulations to assess the accidental load effect due to ship impacts.

Eurocode approach to Ship collision Risk

The Eurocode EN 1991-1-7 (CEN, 2006), which governs accidental actions on structures, provides a probabilistic formulation for evaluating the likelihood of structural failure due to ship collision over a defined design life. The risk of failure is expressed as:

$$P_f(T) = n \times P_c \times T \times (1 - p_a) \times \int P\left(F_{dyn}(x) > R\right) dx \qquad \textit{(Equation 2-6)}$$

Where:

- n = average number of ship passages per year
- P_c = causation probability (i.e., probability that a ship is on a collision course)
- T = design lifetime (typically 100 years)
- p_a = probability of successful evasive action
- $F_{dyn}(x)$= impact force as a function of ship mass and velocity
- R = structural resistance of the impacted member

This equation highlights that failure probability is not only a function of traffic volume, but also of human behavior, ship kinetic energy, and structural capacity. The integral term captures the conditional probability that the impact energy exceeds the structural resistance, taking into account variations in ship characteristics and impact scenarios.

The Eurocode allows engineers to perform risk-informed design by specifying an acceptable failure probability threshold (e.g., $P_f < 10^{-4}$ over a 100-year period). If the estimated risk exceeds this threshold, designers must either increase structural resistance or reduce collision probability through risk mitigation strategies such as VTS, TSS, or barriers.

While this approach is robust for conventional bridges, its direct application to floating bridge structures may require modifications, particularly to reflect dynamic effects such as bridge motion, mooring loads, and impact damping from the surrounding water.

AASHTO approach to ship collision risk

The AASHTO Guide Specification for Vessel Collision Design of Highway Bridges (AASHTO, 2009) adopts a structured risk product model to estimate the probability and severity of accidental ship impacts. The model quantifies the annual expected number of bridge-impact events using the following multiplicative equation:

$$AF = N.P_A.P_G.P_C.P_F \qquad \textit{(Equation 2-7)}$$

Where:

- N = number of vessel passages
- P_A = probability the vessel is on an aberrant path
- P_G = probability the vessel reaches the bridge site

- P_C = probability of contact with the bridge
- P_F = probability of structural failure upon contact

Each factor in this risk product reflects a different stage of the collision progression—starting from traffic characteristics (route geometry and traffic density), moving through failure modes (deviation and loss of control), and ending with structural vulnerability and protection (such as fenders or pier spacing).

This formulation allows for transparent scenario-based risk assessment, facilitating the design of protective measures (e.g., spacing between piers, dolphins, or fender systems). While robust for conventional bridges, the model's direct application to floating structures is limited, as it assumes fixed piers and rigid foundations. For floating or semi-floating bridges, dynamic factors such as hydrodynamic interaction, bridge flexibility, and mooring stiffness would need to be incorporated separately.

2.4.2 Link to Probabilistic Ship Collision Modeling

In this thesis, the collision probability (Pcat1, Pcat2, Pcat3) is computed based on ship traffic patterns, failure modes, and ship behavior using AIS data and KDE-based spatial models. These probabilities directly relate to the likelihood term Pc in the Eurocode expression or the cumulative probability $P_{collision}$ in general design practice.

In performance-based design, the design ship impact load F_d is often taken as:

$$F_d = E.P_{collision} \qquad \textit{(Equation 2-8)}$$

Where:

- $E = \frac{1}{2}m.v^2$ is the kinetic energy of the striking vessel,
- $P_{collision}$ is the probability of such an event occurring during the design life.

Thus, a reduction in collision probability—as achieved through VTS or TSS—allows the designer to adopt lower design impact forces while maintaining safety targets.

Tools like IWRAP MKII and models developed by Pedersen (1995) implement this logic in practice. They estimate annual probability of bridge contact based on geometric ship routing, human failure modes, and environmental effects. These values are then multiplied by kinetic energy terms and adjusted for impact angles and damping to estimate the expected load.

By using a probabilistic approach, designers can:

- Identify the critical ship class (the “design ship”) most likely to govern the load
- Evaluate whether risk-reducing measures justify lower structural demand
- Reduce over-conservatism in regions with traffic control or stable navigation patterns

This shift toward data-informed, probabilistically justified design is especially important for floating structures and offshore bridges, where load conditions vary with traffic direction and environmental exposure.

3 Chapter 3: Methodology

3.1 Overview

This chapter presents the methodology used to estimate ship-bridge collision probabilities for a proposed floating bridge crossing the Nordfjord, between Anda and Lote. The approach builds upon Pedersen’s four-category model, using AIS data and scenario-based simulations to quantify the likelihood of accidental allision events.

The primary objective is to compute the probability of collision—categorized into ordinary route collisions (Cat.1), failed course changes (Cat.2), and drifting/technical failures (Cat.3)—for each ship class and direction of travel. These probabilities are then used to assess the effect of risk-reducing measures (Vessel Traffic Services and Traffic Separation Schemes), and to support the development of performance-based design impact loads for the central pier of the floating bridge.

This methodology directly supports the risk-based design principles outlined in structural codes such as Eurocode and AASHTO, where accidental ship loads are defined as a function of both collision probability and impact energy. By integrating direction-specific traffic patterns, causation factors, and mitigation strategies, the study enables a more realistic and localized estimation of:

- The “design ship” class most likely to govern the load
- The reduction in design force achievable through operational controls
- The effectiveness of VTS/TSS as part of structural optimization

AIS data from October 2023–2024 forms the backbone of the analysis, with kernel density estimation (KDE) and Python-based integration routines used to evaluate probabilistic parameters. This chapter details the step-by-step modeling procedures for each risk category, the implementation of VTS/TSS reductions, and the implications for probabilistic design.

3.2 AIS Data Preparation

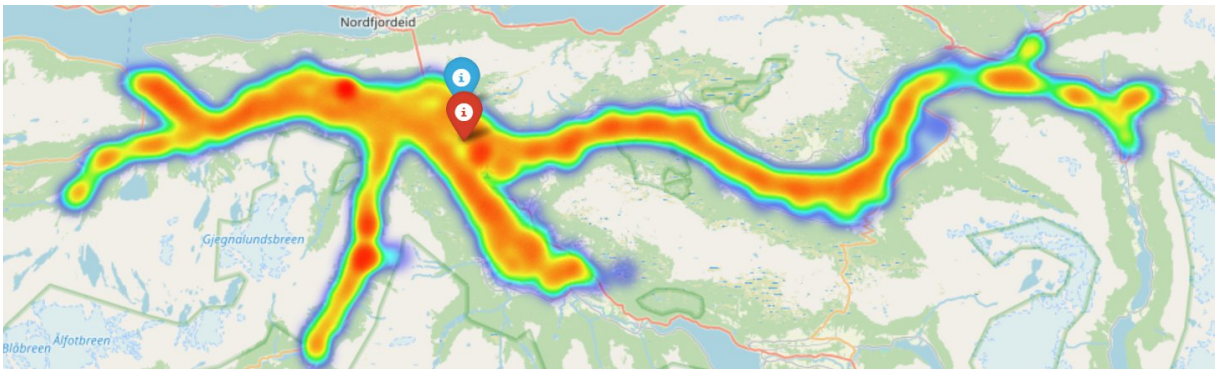


Figure 3-1 Ship Traffic Heatmap

Automatic Identification System (AIS) data was collected in CSV format for the period October 2023 to October 2024, covering vessel movements in a large portion of Nordfjorden. To focus the analysis on ship traffic relevant to the proposed bridge site and reduce computational load, the dataset was geographically constrained to a smaller bounding box:

- Latitude range: 61.81° to 61.915°
- Longitude range: 5.9° to 6.3°

This area includes the Anda–Lote bridge corridor while excluding distant navigational activity unrelated to the bridge crossing risk.

In addition, a number of MMSIs (Maritime Mobile Service Identities) related to passenger ferries were excluded from the dataset:

{41536300,41536301,257017920,257017930,258573000,257268700}

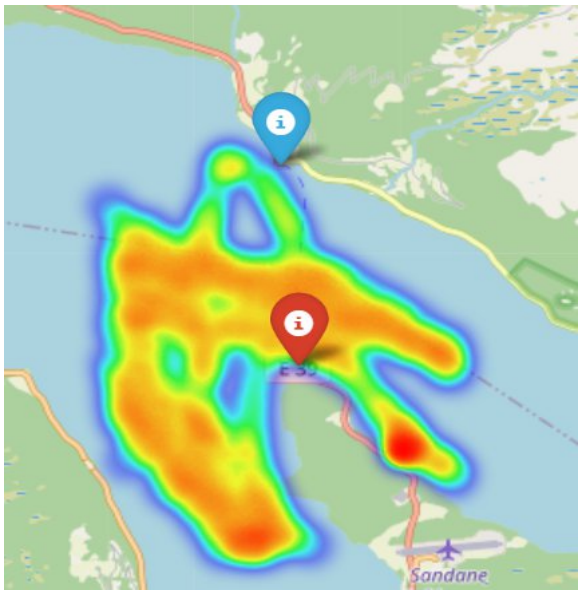


Figure 3-2 Heatmap without Ferries in Bounded Box

These MMSIs correspond to passenger ferries that operate regular crossings in this region. Since the ferry route will no longer be in service after the construction of the bridge, it was appropriate to exclude these vessels from the risk assessment. This ensures that the computed collision probabilities reflect future traffic conditions more realistically.

Following this filtering, the AIS data was cleaned and parsed to extract vessel trajectories. Ship movement was projected onto the Anda–Lote bridge axis to determine lateral crossing positions. By analyzing the trend of longitude values along the projected line, each vessel was classified as eastbound or westbound, which allowed for directional risk modeling in subsequent analysis.

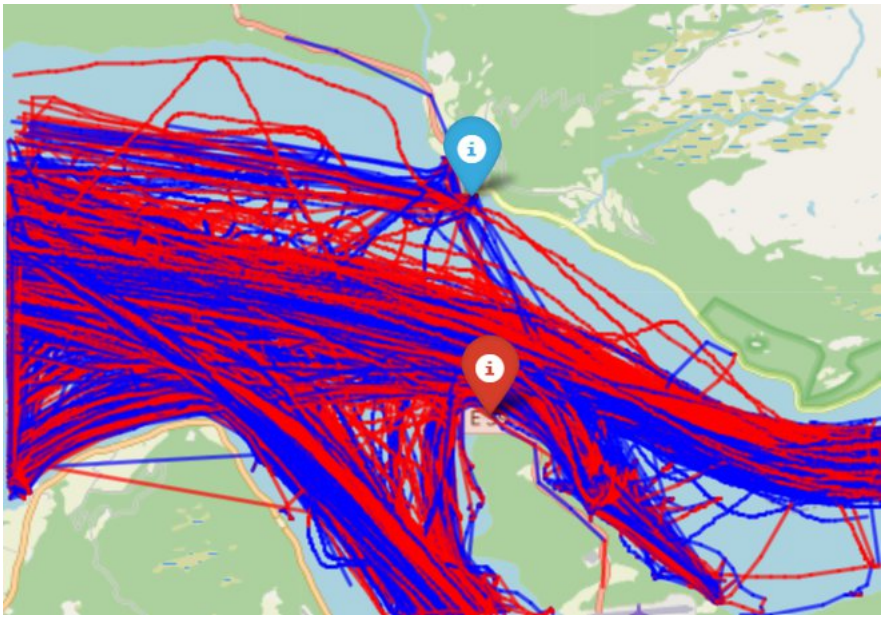


Figure 3-3 Directional Ship Traffic (Westwards, blue, and Eastwards,red)

3.3 Bridge Geometry and Traffic Separation

The bridge alignment was defined as a projected straight line between:

- Anda: (61.84645, 6.08237)
- Lote: (61.86406, 6.07931)

Figure 3-4 Anda and Lote Location

Using UTM projection, ship trajectories were tested for intersection with the bridge axis. The longitudinal intersection points were then used to generate KDEs of crossing density.

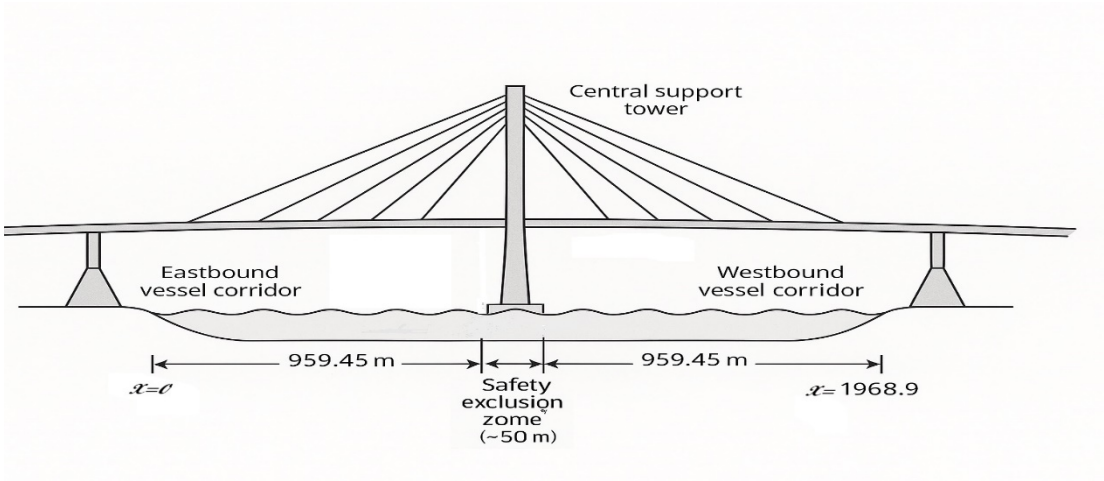


Figure 3-5 Side View of the Bridge

To simplify risk analysis and avoid encounter-based collisions, ship traffic was assumed to follow separate eastbound and westbound lanes, effectively removing Category 4 from the modeling scope.

3.4 Lateral Distribution of Ship Traffic

To estimate the likelihood of bridge allision events (particularly in Category 1 and Category 3), it was necessary to compute the lateral distribution of ship crossings, denoted by f(z), where z is the transverse position along the bridge axis.

Using the filtered AIS data, each ship's movement was projected onto the straight line between Anda and Lote (the proposed bridge alignment). For each track, the intersection point with the bridge axis was computed, and the distance along the axis was stored as the crossing position.

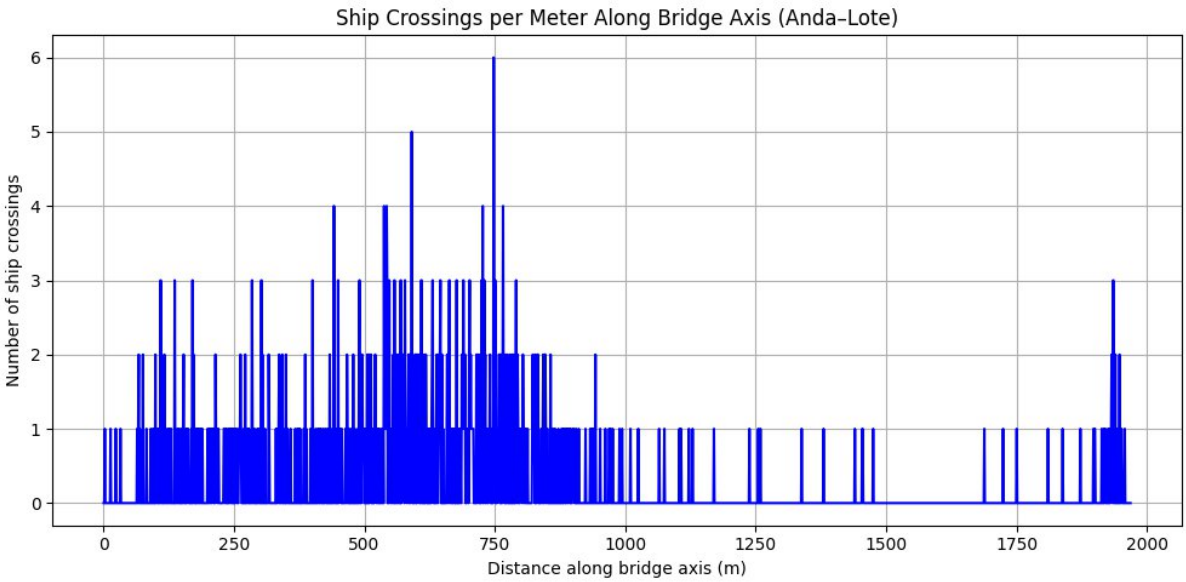


Figure 3-6 Ship Crossings per Meter along Bridge Axis

A Kernel Density Estimation (KDE) technique was then applied to these projected crossing points to obtain a smooth, continuous estimate of the lateral distribution. The KDE was computed using Python's scipy.stats.gaussian_kde function, with a bandwidth of 0.05: see appendix A1 for python code.

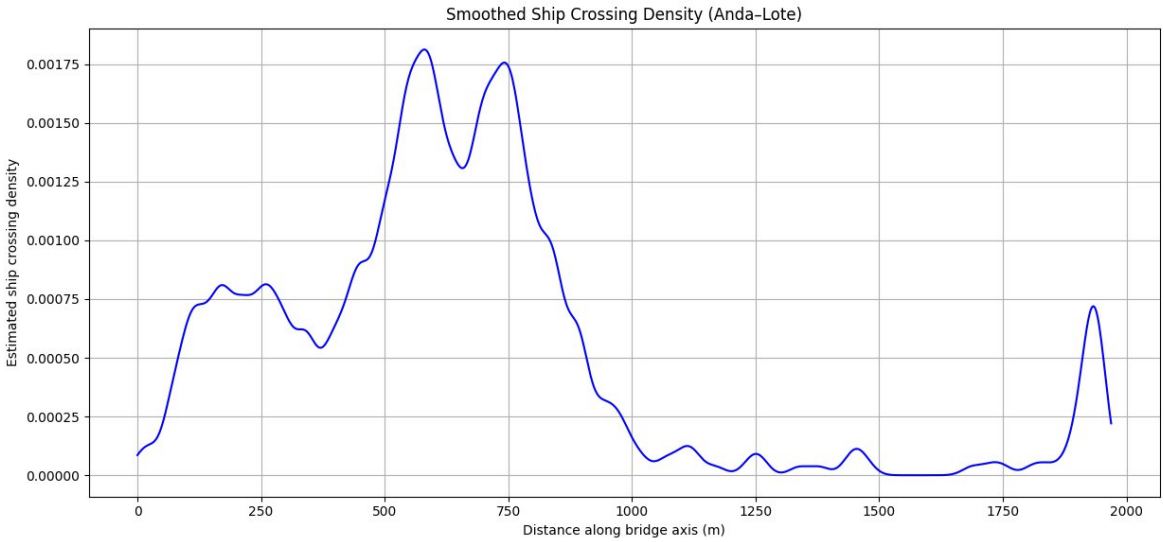


Figure 3-7 Estimated Ship Crossing Density Using KDE

3.4.1 Justification for Using KDE over Parametric Distributions

In many traditional ship collision risk models, the lateral distribution of vessel traffic f(z) is assumed to follow a parametric form, such as a Normal, Lognormal, or Weibull distribution. While convenient, these assumptions impose predefined shapes on traffic patterns that may not accurately reflect the true behavior of vessels in complex waterways.

In this study, a non-parametric Kernel Density Estimation (KDE) method was applied to AIS-derived crossing data to model f(z). This approach does not assume any specific distributional form, allowing the data to speak for itself. KDE captures asymmetries, skewness, and clustering in ship positions that are often observed in real-world traffic—especially in constrained fjord environments where ships may avoid certain areas due to depth, wind exposure, or navigation practices.

Moreover, the use of KDE aligns with modern maritime risk modeling literature, including studies by Goerlandt et al. (2017), Svanberg et al. (2019), and the IWRAP Mk II framework, which all emphasize the importance of empirical data in lateral traffic modeling. By using KDE, this thesis ensures that the calculated collision probabilities are grounded in observed behavior, leading to more realistic and site-specific risk assessments.

3.4.2 KDE Calculation

The KDE was computed using the scipy.stats.gaussian_kde function in Python, which constructs the continuous probability density function f(z) by summing Gaussian kernels centered at each observed data point:

$$f(z) = \frac{1}{nh}\sum_{i=1}^{n} K\left(\frac{z-z_i}{h}\right) \qquad \textit{(Equation 3-1)}$$

Where:

- n is the number of ship crossings
- z_i are the observed crossing positions
- h is the bandwidth (smoothing parameter)
- K is the standard normal kernel function

A bandwidth of 0.05 (normalized to the rescaled channel width) was selected after testing several values for stability and smoothness. This bandwidth controls the balance between smoothness and sensitivity to local features. See Appendix A1.

The resulting f(z) functions were used directly in the double integral formulations for Categories 1–3, where they represent the likelihood of a vessel being in a hazardous lateral position relative to the bridge tower.

3.5 Directional Classification of Ship Traffic

To differentiate between eastbound and westbound ship traffic, each vessel track was analyzed for directional movement. The classification was based on longitudinal trends:

- Vessels showing a consistent increase in longitude along their trajectory were labeled as eastbound
- Vessels with a decreasing longitude were classified as westbound

This method assumes that vessels maintain relatively stable headings within the vicinity of the bridge. Visual inspection of the data, along with analysis of course-over-ground (COG) values, confirmed that this assumption holds true in the study area. The directionally classified trajectories were then assigned to separate KDEs to preserve traffic asymmetries.

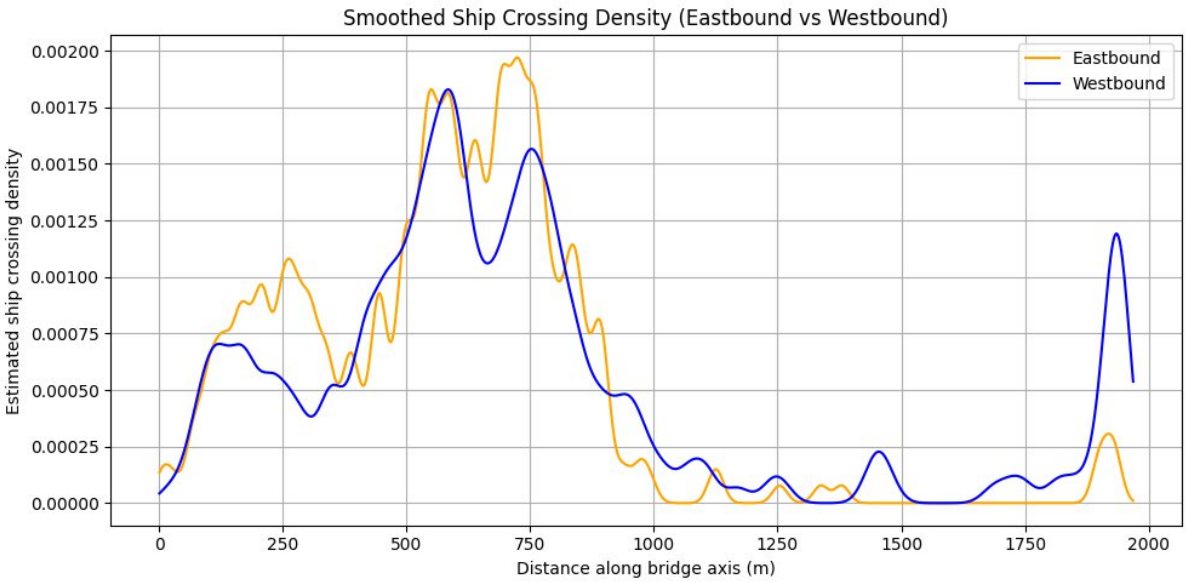


Figure 3.3 Ship Crossing Density by Direction

3.6 Normalization of KDE Domain

To ensure consistency when comparing lateral traffic distributions across different bridge configurations or channel geometries, the KDE outputs were normalized to a fixed spatial domain. The original lateral crossing positions, which span an empirical range of coordinates{ Z_{min}^{orig}, Z_{max}^{orig}} were linearly transformed to a standardized domain { Z_{min}^{new}, Z_{max}^{new}} using the following affine mapping:

$$Z_{new} = \frac{\left(Z_{orig} - Z_{min}^{orig}\right)}{Z_{max}^{orig} - Z_{min}^{orig}} \cdot \left(Z_{max}^{new} - Z_{min}^{new}\right) + Z_{min}^{new} \quad \textit{(Equation 3-2)}$$

This transformation ensures that the KDEs for both directions are mapped onto a common analysis domain. After rescaling, each distribution was renormalized by dividing the KDE by its total integral over the new domain, ensuring the result remains a valid probability density function.

3.7 Geometric Rescaling for Two-Way Traffic Configuration

The proposed bridge includes a central support tower, which bisects the fjord and creates two physically distinct navigation corridors—one for eastbound and one for westbound vessels. To reflect this real-world constraint, the KDEs were rescaled to occupy separate half-width navigation lanes on either side of the tower.

The full lateral span of the observed waterway is approximately 1968.9 meters, with the tower positioned at the midpoint (984.45 m). A safety exclusion zone of ±25

meters was applied, leaving 1918.9 meters of navigable width. This width was evenly split into:

Eastbound domain: $x \in [0, 959.45]$m

Westbound domain: $x \in [1009.45, 1968.9]$m

See figure 3.4.

To map the original KDEs into these new domains, a linear compression was applied using the scale factor:

$$s = \frac{1968.9}{959.45} \approx 2.052$$

Each position x_{old} in the full domain was transformed to a compressed position x_{new} using:

$$x_{new} = \frac{x_{old}}{s} \qquad \textit{(Equation 3-3)}$$

This operation uniformly compressed the KDEs to fit within the half-channel domain while preserving their overall shape, modality, and traffic volume. The result is a pair of symmetrically rescaled PDFs representing the lateral ship traffic density on either side of the bridge tower.

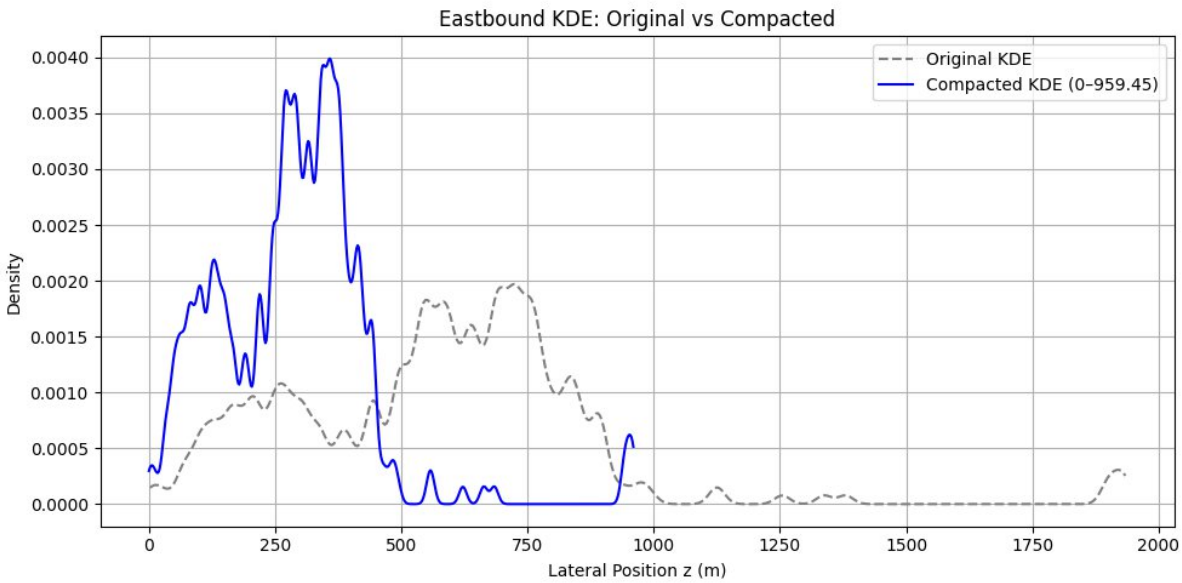


Figure 3.4 KDE Comparison: Original vs Compacted (Eastbound)

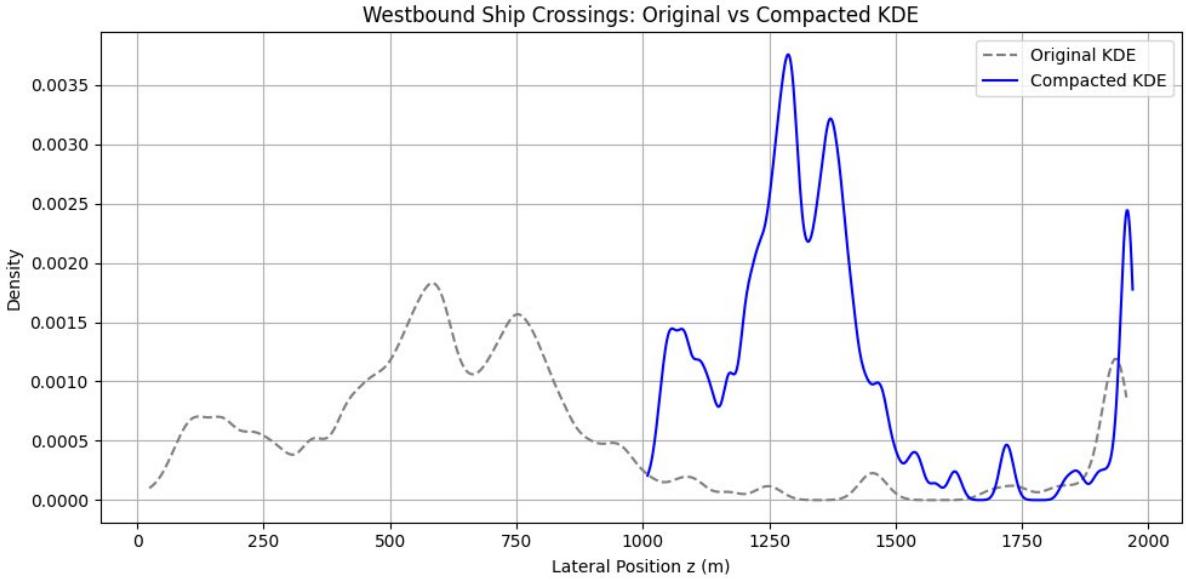


Figure 3.5 KDE Comparison: Original vs Compacted (Westbound)

3.8 Risk Model Implementation

Each of the remaining three Pedersen categories was implemented in Python using numerical integration, KDE functions, and ship class data. The ship population was divided into ten LOA-based classes with traffic volumes sourced from AIS-derived counts.

Table 3-1 Overview of Ship Count by Length Class

LOA (m)	Number of ships
100-125	5
125-150	15
150-175	5
175-200	5
200-225	5
225-250	5
250-275	5
275-300	10
300-325	15
325-350	30

3.8.1 Category 1 — Ordinary Route Collisions

Category 1 collisions represent scenarios in which a ship traveling along its intended sailing route deviates laterally, either due to human error or environmental influence, and collides with the bridge. This type of collision does not involve failed navigational decisions or system faults—it assumes the vessel is proceeding normally, but with a slight offset from its central track that places it at risk of allision with bridge structures.

Kernel Density Estimation of Lateral Distributions

To quantify this risk, the lateral distribution of ship crossings near the bridge axis was modeled using Kernel Density Estimation (KDE). The KDE method estimates the probability density function f(z), where z is the lateral position across the channel. The estimation was performed separately for eastbound and westbound traffic using the gaussian_kde function from Python's scipy.stats module, with a bandwidth of 0.05. This choice ensures a smooth yet responsive density curve suitable for probabilistic integration.

The KDEs were calculated using ship crossing data projected onto the Anda–Lote bridge axis, following domain rescaling procedures described in earlier sections. The KDEs were normalized such that the area under each curve equals one, maintaining valid probabilistic interpretations.

Numerical Integration for Collision Probability

Using the computed KDEs, the probability of Category 1 collision was evaluated via numerical integration over the navigable corridor adjacent to the bridge. For each ship class i, the probability of allision was calculated using equation(2.2):

$$P_{Cat.1} = \frac{1}{T}\sum_{ship\ class\ i}^{n} \int_{Z=Z_a}^{Z_b} \int_0^T P_{c,i}\, Q_i(t)\ f(z)\, dt\, dz$$ (repeated from equation 2.2)

Where:

- T = observation period (1 year)
- $P_{c,i}$ =0.0001 causation factor for class i (assumed constant across all classes)
- $Q_i(t)$= time-dependent ship traffic volume for class i
- $f(z)$= lateral KDE density for direction of travel
- $[Z_a, Z_b]$= bounds of navigable corridor per direction

This double integration accounts for both the spatial probability of being in a hazardous position f(z) and the temporal exposure over the course of the year. The summation accumulates risk across all defined ship length classes (e.g., 100–125 m, 125–150 m, etc.).

Bridge Span Segmentation by Direction

To reflect the real-world bridge geometry and the rescaled channel domain, different integration bounds were applied for each direction:

Westbound: 1009.45–1968.9 m

Eastbound: 0–959.45 m

See Appendix A2 for python scripts

.

3.8.2 Category 2 Modeling and COG-Based Detection of Turning Behavior

Category 2 collisions are defined as events where a vessel fails to perform a required course change, typically at a navigational turning point or constricted area near the bridge. These failures are particularly relevant for bridge configurations involving a central pier or realigned sailing routes.

COG-Based Detection Attempt

To empirically assess the likelihood of failed course changes, Course Over Ground (COG) values from AIS data were analyzed along a ±4–5 km corridor centered on the planned bridge axis. The objective was to detect significant heading changes—typically greater than 10°–15°—that would indicate active navigational correction, such as turning to avoid a central pier.

A Python script (see Appendix A5) was implemented to compute consecutive COG differences along vessel tracks. The heading-change analysis, however, yielded no statistically significant turns across the observed dataset. This suggests:

- Vessels maintain straight, stable headings in the region
- The navigational corridor is geometrically simple, requiring no sharp turns
- The AIS data's temporal resolution may not capture small, gradual corrections

These findings align with similar studies in Nordic fjords, where automated navigation and constrained waterway width often result in very steady course lines (Svanberg et al., 2019). Consequently, a data-driven estimation of turning failures could not be achieved.

Pedersen's Omission Model Implementation

Due to the absence of observed heading shifts, the omission probability P_0 was instead computed using Pedersen's causation model:

$$P_0 = P_t + (1 - P_t) \times P_d^{\frac{d-a}{a}} \qquad \text{(repeated from equation 2.4)}$$

Where:

- P_t=0.01: Probability of complete neglect to alter course
- P_d=0.1: Probability of failing to verify vessel position
- d =500m: Distance sailed on the wrong heading
- a =1000 m: Average navigator attention interval

These values were selected based on conservative assumptions from existing literature (Pedersen, 1995; Friis-Hansen & Simonsen, 2002).

Directional KDE and risk computation

Lateral ship distributions f(z) were computed via Kernel Density Estimation (KDE) of ship intersection points along the bridge axis. These distributions were then used to integrate Category 2 risk across each ship class and traffic direction:

$$P_{Cat.2} = \frac{1}{T}\sum_{ship\ class\ i}^{n} \int_A^B \int_0^T P_0\, P_{c,i}\, Q_i(t)\ f(z)\, dt\, dz \qquad \text{(repeated from equation 2.3)}$$

Where:

- Q_i= number of ship passages in class iii
- $P_{c,i}$ =0.0001 causation factor for class i (assumed constant across all classes)
- $f(z)$= KDE-based lateral distribution
- $[Z_a, Z_b]$ = navigable channel bounds per direction

Direction-specific risk computations were carried out using the scripts in Appendix A3, which implemented the integral numerically for each LOA-based ship class.

3.8.3 Category 3 — Drifting and Technical Failures

Category 3 collisions account for ship allisions caused by technical failures or environmental drifting, where the vessel becomes unintentionally uncontrolled due to issues such as propulsion loss, rudder malfunction, or steering failure. Such failures, combined with wind or current effects, can cause vessels to drift toward the bridge without the possibility of navigational correction.

Model Parameters and Assumptions

This risk scenario is modeled based on the assumed frequency of technical failure and the vessel's passive trajectory during loss-of-control events. The following input parameters were used:

- P_e=0.001: Probability that evasive or corrective action fails, allowing uncontrolled drift
- P_c=0.0001: Causation factor, or the probability that a drifting ship is on a collision path
- T=1 year: Design reference period
- f(z): KDE-derived lateral crossing distribution, representing spatial traffic density

The value of P_e=0.001 was chosen based on recommendations from existing ship-bridge collision risk models, including the IWRAP Mk II framework (Friis-Hansen & Simonsen, 2002) and Pedersen's probabilistic formulations (Pedersen, 1995), both of which suggest this as a conservative yet realistic upper-bound estimate for drifting incidents in confined waterways. Studies of historical allision cases and probabilistic assessments (e.g., DNV, 2013) also indicate that failure probabilities in the order of 10^{-3} are appropriate for modeling rare but hazardous events in bridge design.

Probability Computation

The total Category 3 allision probability was computed by integrating the drifting probability over the lateral span of the navigable channel, as follows:

$$P_{Cat.3} = \frac{1}{T}.P_{c,i}\,.P_e \sum_{ship\ class\ i}^{n} \int_{Z=Z_a}^{Z_b} \int_0^T Q_i(t)\ f(z)\,dt\,dz \quad \text{(repeated from equation 2.5)}$$

The following input parameters were used:

- P_e=0.001: Probability that evasive or corrective action fails, allowing uncontrolled drift
- P_c=0.0001: Causation factor, or the probability that a drifting ship is on a collision path
- T=1 year: Design reference period
- $f(z)$=KDE-derived lateral crossing distribution, representing spatial traffic density

This formulation accounts for the compound probability of:

1. Experiencing a drift-causing failure
2. Lacking the means to anchor or recover
3. Being positioned such that the drift path intersects the bridge axis

3.9 Tools and Libraries Used in Python Script

The analysis in this study was supported by a suite of Python libraries that facilitated data preprocessing, coordinate transformations, statistical modeling, and visualization. The key libraries used are described below:

1. **Pandas**

- Purpose: Handling and cleaning AIS data from CSV files.
- Usage: Parsing columns like mmsi, latitude, longitude, cog, and date_time_utc; grouping vessels by ID; sorting and filtering rows.

2. **NumPy**

- Purpose: Efficient array computations and numerical operations.
- Usage: Calculating distances, directions, and performing vectorized operations for speed and direction analysis.

3. **SciPy**

- Purpose: Statistical integration and KDE smoothing.
- Usage: scipy.stats.gaussian_kde was used to estimate the lateral ship crossing density function f(z). scipy.integrate.quad was used to numerically integrate probability expressions across bridge domains.

4. **Matplotlib**

- Purpose: Visualization of ship traffic patterns and probability distributions.
- Usage: Plotting histograms of ship crossings, kernel density estimates (KDE), and impact load visualizations.

5. **Shapely**

- Purpose: Geometric computation and spatial analysis.
- Usage: Representing ship tracks as LineString objects, checking intersection with the bridge axis, and extracting crossing locations.

6. **PyProj**

- Purpose: Coordinate transformation from geographic (WGS84) to projected coordinate systems.
- Usage: Projecting AIS coordinates to UTM zone 32N (EPSG:32632) or Web Mercator (EPSG:3857) for accurate distance and direction computation.

7. **Globe and OS**

- Purpose: File management and automated batch processing.
- Usage: Locating all .csv files in a directory containing AIS data; handling file paths across systems.

8. Math

- Purpose: Basic trigonometric and absolute value operations.
- Usage: Used in the heading change detection script to compute COG (Course Over Ground) differences.

This toolchain enabled efficient extraction, projection, filtering, and probabilistic analysis of thousands of ship movements, supporting the risk modeling and design assessments presented in this thesis.

3.10 Inclusion of Risk-Reducing Measures (VTS and TSS)

To evaluate the impact of operational safety systems on ship-bridge collision probabilities, this study incorporated the effect of Vessel Traffic Services (VTS) and Traffic Separation Schemes (TSS). Both measures are endorsed by the International Maritime Organization (IMO) and the International Association of Marine Aids to Navigation and Lighthouse Authorities (IALA) as best practices for managing navigational risk in constrained waterways, high-traffic zones, and areas near fixed infrastructure such as bridges.

1 Purpose and Function of VTS and TSS

- VTS provides real-time monitoring and guidance to vessels within a defined coverage area using radar, AIS, VHF communication, and shore-based traffic coordination. Its objective is to reduce navigational error by improving situational awareness and enforcing recommended actions in the event of deviation, technical failure, or collision risk (IALA, 2016).
- TSS establishes segregated traffic lanes, often with dedicated directions of flow, turning points, and separation zones. This system reduces the likelihood of head-on encounters, sharp course changes, and unexpected maneuvering by clearly structuring ship movements in narrow passages or fjords (IMO, 2002).

2 Quantifying the Risk Reduction Effect

Based on guidance from literature and previous maritime risk studies (notably Pedersen, 1995; Friis-Hansen & Engberg, 2017), the effect of VTS and TSS on

allision probabilities can be quantified using scaling factors applied to the key causation probabilities or omission parameters.

Table 3-2 mitigation effect

Mitigation Measure	Source	Estimated Effectiveness
VTS	Pedersen (1995), IWRAP Mk II	~70% reduction in human errors
TSS	Friis-Hansen & Engberg (2017)	~50% reduction in head-on/crossing conflicts

To simplify implementation, these effects were combined multiplicatively and applied to key parameters in each risk category:

- **Category 1 and 2**:
  - Adjusted causation factor:

$$P_c^{adjusted} = P_c \times 0.3 \times 0.5 = 1.5 \times 10^{-5}$$

  - For Category 2, navigational omission parameters were also reduced:
    $P_t = 0.003 (from\ 0.01)$— reduced probability of total course neglect
    $P_d = 0.05 (from\ 0.10)$— reduced likelihood of failing to verify position

Category 3:

- Adjusted evasive failure probability:

$$P_e^{adjusted} = 0.00015 \text{ (original: 0.001)}$$

These adjusted values were implemented in the probability calculations for all three collision categories, both in the form of modified input parameters and by generating separate probabilistic outcomes for:

- Baseline (unmitigated) and
- Mitigated (with VTS/TSS) conditions.

The effect of these measures was evaluated quantitatively (see Chapter 4) by comparing the total allision probability and category-specific risk contributions across both scenarios.

4 Chapter 4: Results

4.1 Overview

This chapter presents the results of the probabilistic ship-bridge collision analysis for the proposed floating bridge across Nordfjord. The analysis follows Pedersen's classification of collision scenarios—Category 1 (Ordinary route collisions), Category 2 (Failed course changes), and Category 3 (Technical failure or drifting).

Although ship traffic was initially classified into eastbound and westbound streams, the resulting collision probabilities for both directions were found to be effectively identical. This is primarily because the study area lies within a dead-end fjord, where every ship entering westbound must eventually exit eastbound, and vice versa. This inherent symmetry in traffic movement ensures that both directions experience identical traffic volumes and distributions over time.

Additionally, KDE-based lateral distributions $f(z)$ were symmetrically rescaled for each side of the bridge (see section 3.7), and the same causation parameters were applied across both flows. Therefore, to avoid redundancy, detailed probability calculations and plots are presented only for westbound traffic. These results apply equally to the eastbound direction.

4.2 Category 1: Ordinary Route Collisions

This category includes vessels traveling along their planned navigation path but deviating laterally toward the bridge due to human error. Probabilities were calculated by integrating AIS-based KDE functions over the navigable channel width, using a causation factor of $P_c = 0.0001$ and a KDE bandwidth of 0.05.

4.2.1 Results by Ship Class (Westbound)

As shown in the corresponding figure, large vessels—particularly those over 300 m in length—contribute the most to risk. The total probability for Category 1 collisions without mitigation is approximately 0.01, which is reduced by half to 0.0015 when VTS/TSS is applied.

Table 4-1 Category 1 Collision Probabilities by Ship Class, mitigated

Ship Length Class (m)	Collision Probability
100–125	0.000075

Ship Length Class (m)	Collision Probability
125–150	0.000225
150–175	0.000075
175–200	0.000075
200–225	0.000075
225–250	0.000075
250–275	0.000075
275–300	0.000150
300–325	0.000225
325–350	0.000450

Table 4-2 Category 1 Collision Probabilities by Ship Class, non-mitigated

Ship Length Class (m)	Collision Probability
100–125	0.000500
125–150	0.001500
150–175	0.000500
175–200	0.000500
200–225	0.000500
225–250	0.000500
250–275	0.000500
275–300	0.001000
300–325	0.001500
325–350	0.003000

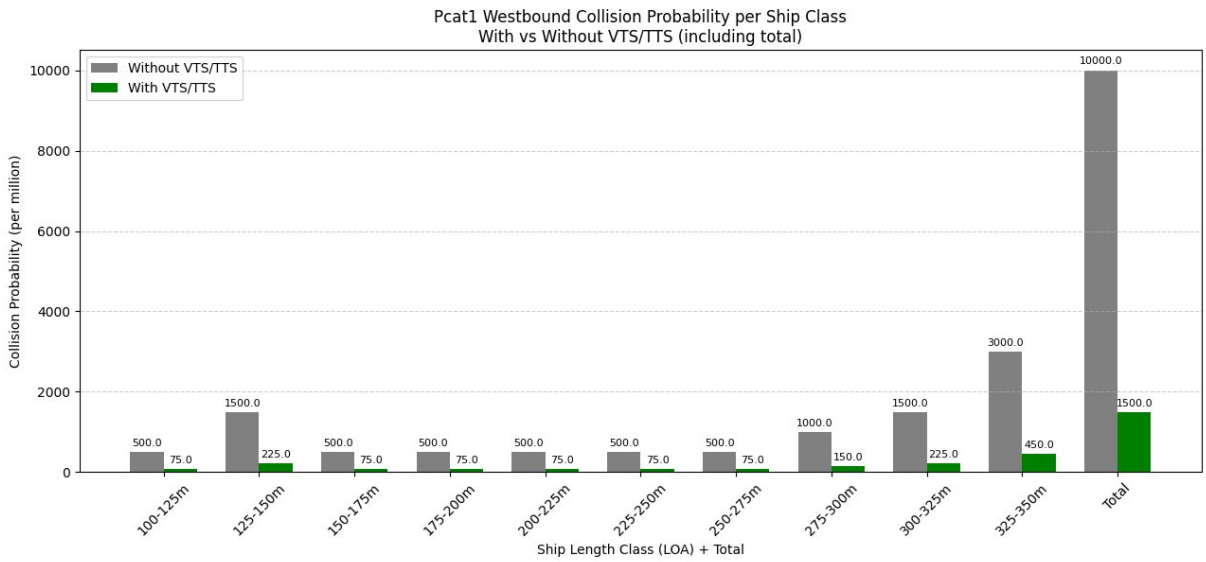


Figure 4-1Category 1 Probabilities (Westbound with/without VTS/TSS)

Because of the symmetric traffic flow (due to the fjord's dead-end nature), these values also hold for eastbound ships and are not repeated separately.

4.3 Category 2: Failed Course Change

Category 2 includes ships that fail to alter their heading near the bridge due to human navigational error. The lateral KDE distribution $f(z)$ was derived from AIS data, while P_0 was computed using Pedersen's model, as detailed in section 3.8.2.

An analysis of heading data near the bridge axis revealed no significant course alterations (>15°) in the ±4–5 km corridor, suggesting that turning behavior is not pronounced in the area or is smoothed over long distances. As a result, assumed values for navigator check intervals and deviation distances were used in the P_0 formulation.

4.3.1 Results by Ship Class (Westbound)

The westbound results show that Category 2 contributes the largest share of total collision risk, with a value of approximately 0.03140655 without mitigation, and 0.00669258 with VTS/TSS—a 78% reduction. Given the equal number of inbound and outbound voyages in this closed-end fjord, these values also apply to eastbound traffic.

Table 4-3 Category 2 Collision Probabilities by Ship Class, mitigated

Ship Length Class (m)	Collision Probability
100–125	0.00033463
125–150	0.0010839
150–175	0.00033463
175–200	0.00033463
200–225	0.00033463
225–250	0.00033463
250–275	0.00033463
275–300	0.00066926
300–325	0.00100389
325–350	0.00200777

Table 4-4 Category 2 Collision Probabilities by Ship Class, non-mitigated

Ship Length Class (m)	Collision Probability
100–125	0.00157033
125–150	0.00471098
150–175	0.00157033
175–200	0.00157033
200–225	0.00157033
225–250	0.00157033
250–275	0.00157033
275–300	0.00314065
300–325	0.00471098
325–350	0.00942196

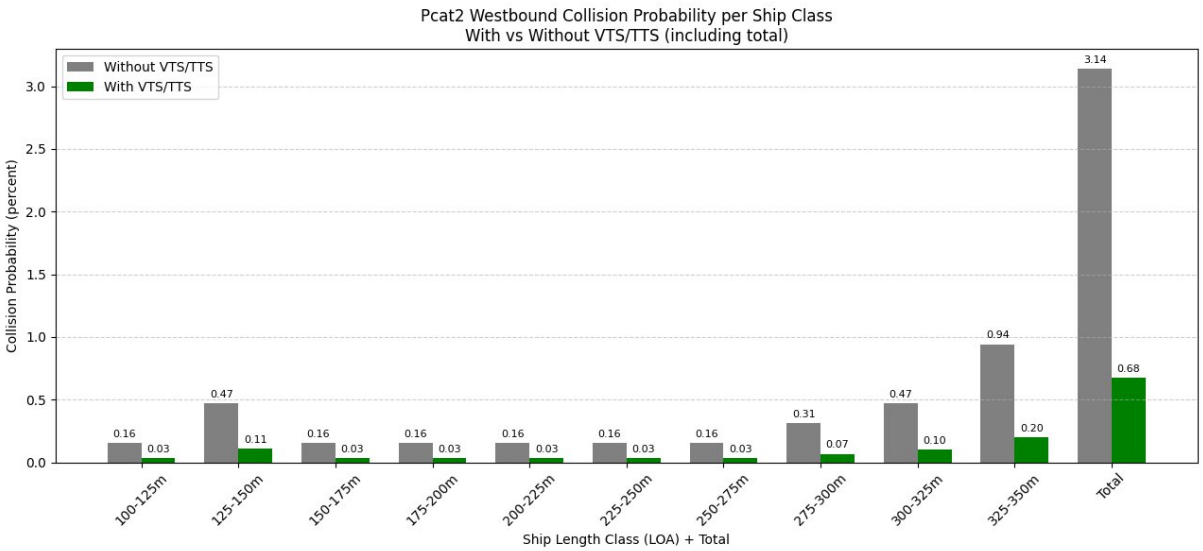


Figure 4-2Category 2 Probabilities (Westbound with/without VTS/TSS)

4.4 Category 3: Evasive Maneuver and Drifting

Category 3 accounts for technical failures or evasive actions that result in unintended bridge collisions. Given the lack of contextual maneuvering data in AIS, this was modeled using a fixed probability $P_e = 10^{-3}$ as discussed in Methodology (see section 3.10). This was adjusted to 1.5×10^{-5} under VTS/TSS conditions.

4.4.1 Results by Ship Class (Westbound)

This category exhibits very low total probabilities—on the order of 1.0×10^{-5} without mitigation and 2.248424×10^{-7} with VTS/TSS. Again, because all ships must eventually pass both directions in and out of the fjord, these values apply symmetrically to eastbound traffic.

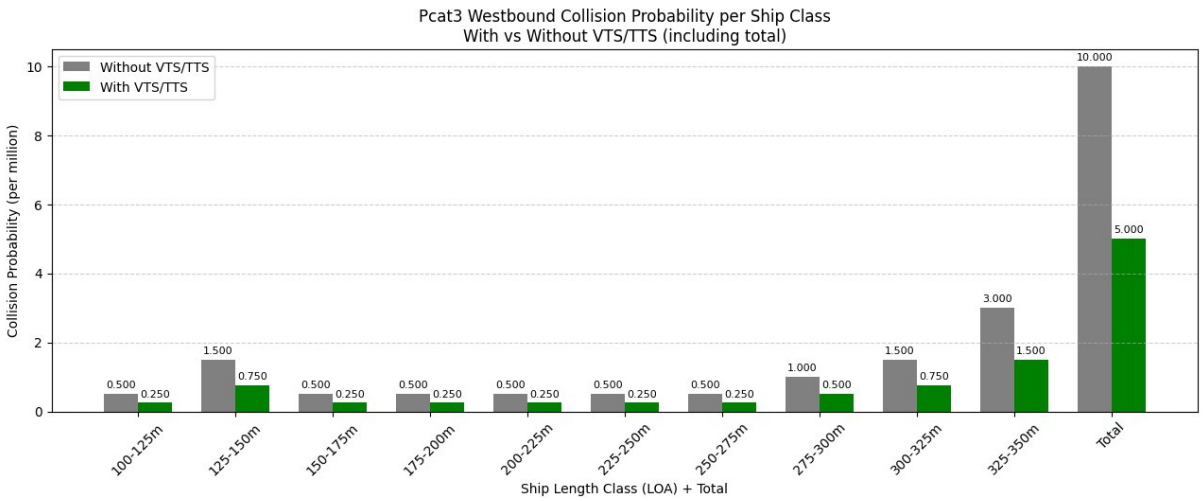
Table 4-5 Category 2 Collision Probabilities by Ship Class, mitigated

Ship Length Class (m)	Collision Probability
100–125	1.122694×10^{-8}
125–150	3.375693×10^{-8}
150–175	1.122694×10^{-8}
175–200	1.122694×10^{-8}
200–225	1.122694×10^{-8}

Ship Length Class (m)	Collision Probability
225–250	1.122694×10^{-8}
250–275	1.122694×10^{-8}
275–300	2.245388×10^{-8}
300–325	3.375693×10^{-8}
325–350	6.751303×10^{-8}

Tabell 4-6 Category 2 Collision Probabilities by Ship Class, non-mitigated

Ship Length Class (m)	Collision Probability
100–125	5.000000×10^{-7}
125–150	1.500000×10^{-6}
150–175	5.000000×10^{-7}
175–200	5.000000×10^{-7}
200–225	5.000000×10^{-7}
225–250	5.000000×10^{-7}
250–275	5.000000×10^{-7}
275–300	1.000000×10^{-6}
300–325	1.500000×10^{-6}
325–350	3.000000×10^{-6}



Figur 4-3Category 3 Probabilities (Westbound with/without VTS/TSS)

4.5 Summary of Collision Risk by Category

Table 4-7 Summary of collision probabilities without and with mitigation

Scenario	P_cat1	P_cat2	P_cat3	Total
Without VTS/TSS	0.010000	0.03140655	0.00001	41.41655×10^{-3}
With VTS/TSS	1.5×10^{-3}	6.69258×10^{-3}	2.248424×10^{-7}	8.19280484×10^{-3}

This table confirms that:

- Category 2 poses the highest allision risk across all ship classes
- VTS and TSS measures reduce collision probabilities by approximately 80%
- Eastbound and westbound risks are equal due to symmetric traffic volumes, mirrored KDE domains, and the natural flow pattern of the dead-end fjord

As a result, the westbound results shown throughout this chapter are representative of the entire traffic system near the bridge.

5 Chapter 5: Design Ship Selection and Impact Load Assessment

5.1 Selection of Design Ship

In accordance with Eurocode EN 1991-1-7 and Pedersen's methodology, the design ship for impact analysis is selected based on the vessel class that contributes the highest probability-weighted impact energy to the total collision risk. This combines both the frequency of occurrence and the energy transferred during a potential impact.

Based on the probabilistic results presented in Chapter 4, vessels in the 325–350 m LOA class were found to contribute most significantly to total collision probability, particularly in Category 2, which dominates the risk profile. Therefore, a representative vessel of this class was selected as the design ship.

Assumed parameters for this class (based on IACS and DNV-GL references):

Table 5-1 Design ship parameters used for impact load calculation.

Parameter	Value
Length overall (LOA)	340 m
Beam (B)	50 m
Draft (T)	15 m
Displacement (Δ)	150,000 tonnes
Service speed (V)	8.0 m/s (approx. 15.6 knots)
Impact velocity (V_n)	3.0 m/s (assumed at impact)
Added mass coefficient (k_a)	1.2 (normal to bow direction)

5.1.1 Impact Load Calculation

The dynamic impact force is evaluated using a simplified energy-based model, assuming full kinetic energy transfer over a deformation length:

$$E_k = \frac{1}{2}.m_{eff}.V_n^2 \qquad \textit{(Equation 5-1)}$$

$$m_{eff} = k_a.m = 1.2 \times 150000 \times 10^3 = 180 \times 10^6 kg$$

$$E_k = \frac{1}{2} \times 180 \times 10^6 \times 3.0^2 = 810 \times 10^6 J = 810 MJ$$

Assuming the energy is absorbed over a deformation length of 2 m:

$$F_{impact} = \frac{E_k}{d} \qquad \textit{(Equation 5-2)}$$

$$F_{impact} = \frac{810 \times 10^6}{2} = 405 \times 10^6 = 405 MN$$

This represents the maximum theoretical force during a rigid impact. In practice, this is compared against bridge pier resistance (R) to assess risk acceptability.

5.1.2 Adjusted Risk-Based Impact Force

In risk-informed design, the design impact load is often scaled based on the probability of collision over the design life TTT. Using the total collision probability from Chapter 4:

- Without VTS/TSS: $P_{col} = 41.41655 \times 10^{-3}$
- With VTS/TSS: $P_{col} = 8.19280 \times 10^{-3}$

Then the probabilistic design impact (expected impact load) is approximated as:

$$F_{expexted} = P_{col} \times F_{impact} \qquad \textit{(Equation 5-3)}$$

Table 5-2 Expected impact load with and without mitigation

Scenario	P_{col}	$F_{expexted}$
Without VTS/TSS	41.41655×10^{-3}	16.77370
With VTS/TSS	8.19280×10^{-3}	3.31808

This comparison shows that implementing VTS and TSS reduced the expected impact load approximately 80%, making a strong case for considering such measures in the design phase.

5.1.3 Implications for Structural Design

The assessment of ship-bridge collision risk is not complete without translating the results into practical design actions. In this study, the design ship—representing the vessel class with the highest risk-weighted contribution—was found to produce an estimated maximum impact force of 405 MN, assuming full energy transfer in a rigid impact scenario. This value forms the basis for understanding how the bridge structure, particularly the mid-span tower, must perform under extreme accidental loading conditions.

Unmitigated Scenario: Full Risk Envelope

In the absence of any risk-reducing measures (e.g., VTS or TSS), the computed total probability of collision was approximately 4.1% over the design life of the structure (typically 100 years). This means that, under unmitigated conditions, there is a non-negligible chance that a vessel may impact the bridge with a force on the order of 405 MN.

Such a high design force has several implications:

- Material and structural demands on the tower increase significantly, possibly requiring reinforced piers, larger foundations, or protective fender systems.
- The bridge must be designed either to absorb this energy without collapse, or to redirect it through sacrificial structural components.
- The design must consider the structural response under dynamic impact, including stress wave propagation, plastic deformation, and stability after impact.

If the structure is designed strictly for this 405 MN load with 4.1% occurrence, the design may become costly, conservative, or structurally over-dimensioned.

Mitigated Scenario: Design Optimization under Reduced Risk

When VTS and TSS are implemented, the total probability of collision drops to approximately 0.8%. From a design perspective, this allows for a probabilistic adjustment of the impact force by incorporating expected values, rather than relying on worst-case outcomes alone.

The risk-weighted expected force is reduced to:

$$F_{expexted} = 8.19280 \times 10^{-3} \times 405MN = 3.31808MN$$

This has several key advantages:

- The structural system can be optimized for lower accidental loads, reducing material use and construction cost.
- The design may allow for more slender or flexible tower geometry, which is particularly important for floating bridges where weight and buoyancy balance are critical.
- Maintenance and inspection intervals can be adjusted based on lower fatigue demand from rare events.

However, the designer must still ensure that the probabilistic reliability of the system is acceptable under regulatory or client-defined risk criteria.

Target Reliability and Probabilistic Design Verification

Designing for accidental actions in a probabilistic context requires the use of target reliability indices. For rare events like ship impact, a failure probability of $P_f < 10^{-4}$ over the design life is commonly cited in literature and partial factor approaches (Eurocode EN 1990 Annex B).

To confirm that the structure satisfies this threshold, one of the following approaches must be used:

Full probabilistic analysis using Monte Carlo or FORM/SORM to compare the load distribution F_{impact} with structural resistance R, and verify $P(F_{impact} > R) < 10^{-4}$.

Partial factor method, where design values are obtained by applying safety factors to both load and resistance:

$$\gamma_F . F_{design} < \frac{R_k}{\gamma_M} \qquad \textit{(Equation 5-4)}$$

where γ-factors calibrated to achieve the desired reliability.

This ensures that the actual design force (whether full or reduced) is compatible with the structure's probabilistic resistance, and the global safety margin is maintained.

The analysis confirms that risk-reducing measures such as VTS/TSS do not just improve operational safety—they also provide a quantitative basis for optimizing structural design. By reducing collision probabilities, they enable more efficient design solutions without compromising reliability. However, adopting lower impact loads must be accompanied by rigorous reliability checks, in accordance with probabilistic safety formats or partial factor design procedures as outlined in Eurocode.

6 Chapter 6: Conclusion and Recommendation

6.1 Conclusion

This study has developed and implemented a data-driven framework for evaluating the risk of ship collisions with a proposed floating bridge across the Nordfjord. By combining Automatic Identification System (AIS) data with Pedersen's probabilistic categorization of allision scenarios, the thesis provides both a conceptual and quantitative assessment of collision risks. The analysis encompassed three principal categories of collision scenarios—Category 1 (ordinary route deviations), Category 2 (failed course changes), and Category 3 (drifting due to technical failure)—with the unique addition of risk mitigation through Vessel Traffic Services (VTS) and Traffic Separation Schemes (TSS).

One of the core contributions of this thesis is the use of realistic lateral traffic distributions, estimated via Kernel Density Estimation (KDE) directly from AIS crossing data. This approach allows the spatial behavior of vessel traffic near the bridge to be accurately modeled without reliance on idealized (e.g., Normal or Weibull) assumptions. Furthermore, AIS-based estimation of impact velocities at the point of crossing enables a more realistic assessment of kinetic energy transfer during a potential impact, improving upon prior assumptions of constant or worst-case speeds.

The design ship was selected as the vessel class with the highest risk-weighted contribution to total allision probability. For the Nordfjord site, this was found to be large vessels in the 325–350 m LOA range, with most of their contribution arising from Category 2 scenarios. The calculated maximum impact force of approximately 405 MN, derived from rigid energy transfer assumptions, represents the upper limit of design loading in the absence of mitigation. However, when scaled by the total probability of collision over a 100-year design life, the expected design force drops to 16.8 MN, and further to 3.32 MN with VTS and TSS in place.

Importantly, the symmetrical layout of the fjord and the separation of traffic lanes into eastbound and westbound channels ensures that each vessel passing westward will also pass eastward, and vice versa. As a result, the statistical distributions of crossing traffic, as well as the resulting probabilities of collision, were shown to be

effectively identical in both directions—allowing the westbound risk to represent the total risk scenario with sufficient accuracy.

This study confirms that VTS and TSS are not only operational safeguards but also powerful tools for design optimization. Their incorporation enables designers to reduce structural redundancy, material use, and costs, while maintaining required safety levels. These findings align with the evolving approach in structural engineering toward performance-based and risk-informed design, as recommended in Eurocode EN 1990 and other modern standards.

6.2 Recommendations for Future Research and Design Practices

While the methodology presented here represents a significant step toward more realistic and integrated bridge impact assessment, there are several opportunities for further development and refinement:

1. **Improved AIS Data Resolution**

AIS messages used in this study had limited temporal resolution (e.g., 1–10 minutes between messages), which may obscure subtle maneuvers or short-term heading adjustments near the bridge. Future analyses could benefit from:

- Higher-frequency AIS data (e.g., from shore-based receivers or port authorities)
- Integration with radar tracking or real-time VTS logs to capture micro-navigation patterns

2. **Full Inclusion of Category 4**

Category 4 collisions—caused by evasive maneuvers or traffic encounters—were excluded due to the enforced directional traffic separation across the bridge. In more complex environments (e.g., open channels or near turning basins), simulating ship domain overlap and encounter rates using dynamic modeling (e.g., agent-based or multi-ship simulations) would enhance completeness.

3. **Structural Response Simulation**

The current impact model is energy-based and assumes full energy transfer into the bridge structure. To assess the structural performance and resilience, further steps include:

- Finite Element Analysis (FEA) of the bridge tower or foundation (e.g., LS-DYNA, ABAQUS)
- Dynamic time-history simulations of ship-structure interaction
- Evaluation of plastic deformation, local crushing, and post-impact recovery

4. **Environmental and Operational Factors**

Drifting scenarios (Category 3) can be heavily influenced by environmental conditions such as:

- Wind speed and direction
- Current velocity
- Emergency anchoring success rates

Incorporating weather and oceanographic datasets would allow a more realistic probability distribution of uncontrolled drift toward the bridge.

5. Probabilistic Design Integration

To fully align with Eurocode EN 1990 Annex B, the structural design process should consider:

- Target reliability index (e.g., β>3.8, corresponding to $P_f < 10^{-4}$)
- Use of FORM/SORM or Monte Carlo methods to compare the statistical distribution of impact loads with structural resistance
- Calibration of partial safety factors for accidental actions specific to ship-bridge impact scenarios

6. Real-Time Risk Monitoring Tools

Beyond design-phase applications, the methodologies developed here could inform operational risk dashboards for bridge authorities. These tools could:

- Use live AIS feeds to compute real-time P(cat) probabilities
- Alert operators when a vessel deviates from expected routes
- Support dynamic bridge opening protocols or traffic advisories

6.3 Final Remarks

The thesis demonstrates that the integration of probabilistic modeling, real-world AIS data, and risk-reduction measures leads to a more efficient and robust bridge design process. The shift from worst-case deterministic assumptions to risk-calibrated design forces represents a major advancement in the structural engineering of critical infrastructure.

By leveraging high-resolution maritime traffic data and adapting global safety practices to local fjord conditions, the approach outlined in this study offers a replicable model for floating bridge projects worldwide—particularly in environmentally sensitive or topographically constrained regions such as coastal Norway.

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8 Appendix A: Python Script

8.1 A.1 KDE Estimation – Shipcrossing(count & estimated density).py

```
import pandas as pd
import numpy as np
import os
import glob
from shapely.geometry import LineString
from pyproj import Transformer
import matplotlib.pyplot as plt
from scipy.stats import gaussian_kde

# === Configuration ===
folder_path = r"C:\path\to\AIS\data "
excluded_mmsis = {41536300, 41536301, 257017920, 257017930, 258573000, 257268700}
bin_size = 1.0 # meters

# === Anda and Lote coordinates in (latitude, longitude) ===
anda_lat, anda_lon = 61.84645, 6.08237
lote_lat, lote_lon = 61.86406, 6.07931

# === Coordinate transformation: WGS84 to UTM Zone 32N (accurate for Norway)
transformer = Transformer.from_crs("EPSG:4326", "EPSG:32632", always_xy=True)
anda_proj = transformer.transform(anda_lon, anda_lat) # (lon, lat)
lote_proj = transformer.transform(lote_lon, lote_lat)

# === Bridge axis and length ===
bridge_line = LineString([anda_proj, lote_proj])
bridge_length = bridge_line.length
print(f"Projected bridge length: {bridge_length:.2f} meters")

# === Load and clean AIS data ===
files = glob.glob(os.path.join(folder_path, "*.csv"))
```

```
df_list = []

for file in files:
  try:
    df = pd.read_csv(file, delimiter="\t", encoding="utf-8", on_bad_lines='skip')
    df.columns = [col.lower().strip() for col in df.columns]

    required = ['mmsi', 'latitude', 'longitude', 'date_time_utc']
    if all(col in df.columns for col in required):
      df = df.dropna(subset=required)
      df['mmsi'] = pd.to_numeric(df['mmsi'], errors='coerce')
      df = df[~df['mmsi'].isin(excluded_mmsis)]
      df['date_time_utc'] = pd.to_datetime(df['date_time_utc'], errors='coerce')
      df = df.sort_values(by=['mmsi', 'date_time_utc'])
      df_list.append(df)
    else:
      print(f"Skipping {file} — missing required columns.")
  except Exception as e:
    print(f"Error reading {file}: {e}")

# === Combine all cleaned data ===
df = pd.concat(df_list, ignore_index=True)

# === Generate ship LineStrings ===
ship_lines = []
for mmsi, group in df.groupby('mmsi'):
  if len(group) < 2:
    continue
  coords = [transformer.transform(lon, lat) for lon, lat in zip(group['longitude'], group['latitude'])]
  ship_lines.append(LineString(coords))

# === Project all crossing distances for KDE and histogram ===
crossing_distances = []
```

```
for ship in ship_lines:
  if ship.crosses(bridge_line) or ship.intersects(bridge_line):
    intersection = ship.intersection(bridge_line)
    if intersection.geom_type == 'Point':
      dist = bridge_line.project(intersection)
      crossing_distances.append(dist)
    elif intersection.geom_type == 'MultiPoint':
      for pt in intersection.geoms:
        dist = bridge_line.project(pt)
        crossing_distances.append(dist)

# === Histogram (binned counts) ===
bin_edges = np.arange(0, bridge_length + bin_size, bin_size)
hist_counts, _ = np.histogram(crossing_distances, bins=bin_edges)
bin_centers = (bin_edges[:-1] + bin_edges[1:]) / 2

# === KDE ===
kde = gaussian_kde(crossing_distances, bw_method=0.05)
x_vals = np.linspace(0, bridge_length, 500)
kde_vals = kde(x_vals)

# === Plot both ===
plt.figure(figsize=(10, 5))
plt.bar(bin_centers, hist_counts, width=bin_size, alpha=0.4, label='Raw Histogram', color='gray',
edgecolor='black')
plt.plot(x_vals, kde_vals * len(crossing_distances) * bin_size, color='blue', label='Smoothed KDE')

plt.xlabel("Distance along bridge axis (m)")
plt.ylabel("Ship crossings (count & estimated density)")
plt.title("Ship Crossings Along Bridge Axis (Anda–Lote)")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```

8.2 A.2 Category 1 Risk – pcat1_west.py

```
# Pcat1

# Collision Probability by Ship Class (Category 1, Westbound - Compacted KDE)

# use east_crossing for Pcat1East


import numpy as np

import scipy.integrate as integrate

from scipy.stats import gaussian_kde

import matplotlib.pyplot as plt


# === Load precomputed westbound crossing data ===

import sys

sys.path.append(r"C:\path\to\AIS\data ")

from Fz import west_crossings  # All raw crossing positions (uncompacted) # use east_crossing for
Pcat1Eastbound


# === Define domain for compacted KDE ===

z_min_new, z_max_new = 1009.45, 1968.9


# === Step 1: Rescale west_crossings into the new domain ===

old_min, old_max = min(west_crossings), max(west_crossings)


def rescale_z(z, old_min, old_max, new_min=z_min_new, new_max=z_max_new):

    return new_min + ((z - old_min) / (old_max - old_min)) * (new_max - new_min)


compact_crossings = [rescale_z(z, old_min, old_max) for z in west_crossings]


# === Step 2: Compute compacted KDE ===

compact_kde = gaussian_kde(compact_crossings, bw_method=0.05)

f_z_raw = lambda z: compact_kde(z) if z_min_new <= z <= z_max_new else 0


# === Optional Step 3: Normalize the KDE ===

area_under_kde, _ = integrate.quad(f_z_raw, z_min_new, z_max_new)

f_z_kde = lambda z: f_z_raw(z) / area_under_kde if area_under_kde > 0 else 0
```

```
# === Also compute original KDE ===
original_kde = gaussian_kde(west_crossings, bw_method=0.05)
orig_min, orig_max = min(west_crossings), max(west_crossings)
f_z_orig = lambda z: original_kde(z) if orig_min <= z <= orig_max else 0

# === Ship classification by Length Overall (LOA) ===
ship_classes = {
    (100, 125): 5,
    (125, 150): 15,
    (150, 175): 5,
    (175, 200): 5,
    (200, 225): 5,
    (225, 250): 5,
    (250, 275): 5,
    (275, 300): 10,
    (300, 325): 15,
    (325, 350): 30
}

# === Constants ===
T = 1  # Time unit
P_c = 0.0001  # Causation probability
domain_start, domain_end = z_min_new, z_max_new

# === Main computation function ===
def compute_pcat1_west_compacted():
    total_P_cat1 = 0
    results = {}

    for (loa_min, loa_max), num_ships in ship_classes.items():
        Q_i = num_ships / T
        integrand = lambda z: P_c * Q_i * f_z_kde(z) * T
        integral_value, _ = integrate.quad(integrand, domain_start, domain_end)
```

```
        results[f"{loa_min}-{loa_max}m"] = integral_value
        total_P_cat1 += integral_value

    return results, total_P_cat1

# === Plot KDEs ===
def plot_kde():
    z_vals_compact = np.linspace(z_min_new, z_max_new, 500)
    kde_vals_compact = [f_z_kde(z) for z in z_vals_compact]

    z_vals_orig = np.linspace(orig_min, orig_max, 500)
    kde_vals_orig = [f_z_orig(z) for z in z_vals_orig]

    plt.figure(figsize=(10, 5))
    plt.plot(z_vals_orig, kde_vals_orig, label="Original KDE", color="gray", linestyle="--")
    plt.plot(z_vals_compact, kde_vals_compact, label="Compacted KDE", color="blue")
    plt.title("Westbound Ship Crossings: Original vs Compacted KDE")
    plt.xlabel("Lateral Position z (m)")
    plt.ylabel("Density")
    plt.grid(True)
    plt.legend()
    plt.tight_layout()
    plt.show()

# === Run ===
if __name__ == "__main__":
    results, total = compute_pcat1_west_compacted()
    print("Collision Probability by Ship Class (Category 1, Westbound - Compacted KDE):")
    for cls, prob in results.items():
        print(f"{cls}: {prob:.6f}")
    print(f"\nTotal Collision Probability (Westbound): {total:.6f}")

    # Plot both original and compacted KDE shapes
    plot_kde()
```

8.3 A.3 Category 2 Risk – pcat2_west.py

```
#Category 2 Collision Probabilities by Ship Class (Westbound, Compacted Domain)
# use east_crossing for Pcat1Eastbound

import pandas as pd
import numpy as np
import os
import glob
from shapely.geometry import Point, LineString
from pyproj import Transformer
import matplotlib.pyplot as plt
from scipy.stats import gaussian_kde
import scipy.integrate as integrate

# === CONFIGURATION ===
folder_path = r"C:\path\to\AIS\data "
excluded_mmsis = {41536300, 41536301, 257017920, 257017930, 258573000, 257268700}
anda_latlon = (61.84645, 6.08237)
lote_latlon = (61.86406, 6.07931)

# === SHIP CLASSIFICATION ===
ship_classes = {
  (100, 125): 5,
  (125, 150): 15,
  (150, 175): 5,
  (175, 200): 5,
  (200, 225): 5,
  (225, 250): 5,
  (250, 275): 5,
  (275, 300): 10,
  (300, 325): 15,
  (325, 350): 30
```

```
}

T = 1 # Time period
P_c = 0.0001 # Causation factor
P_t = 0.01
P_d = 0.1
d = 500 # Distance sailed wrongly
a = 1000 # Position check interval

def compute_P0(d, a):
  return P_t + (1 - P_t) * (P_d ** ((d - a) / a))

# === Coordinate transform and bridge line ===
transformer = Transformer.from_crs("epsg:4326", "epsg:3857", always_xy=True)
anda_proj = transformer.transform(anda_latlon[1], anda_latlon[0])
lote_proj = transformer.transform(lote_latlon[1], lote_latlon[0])
bridge_line = LineString([anda_proj, lote_proj])
bridge_length = bridge_line.length
bridge_vector = np.array([lote_proj[0] - anda_proj[0], lote_proj[1] - anda_proj[1]])

# === Collect crossing positions ===
west_proj_dists = []
csv_files = glob.glob(os.path.join(folder_path, "*.csv"))
for file in csv_files:
  try:
    df = pd.read_csv(file, delimiter="\t", encoding="utf-8", on_bad_lines='skip')
    df.columns = [col.lower().strip() for col in df.columns]
    df = df.dropna(subset=['mmsi', 'latitude', 'longitude', 'date_time_utc'])
    df['mmsi'] = pd.to_numeric(df['mmsi'], errors='coerce')
    df = df[~df['mmsi'].isin(excluded_mmsis)]
    df['date_time_utc'] = pd.to_datetime(df['date_time_utc'], errors='coerce')
    df = df[
      (df['latitude'].between(61.81, 61.915)) &
      (df['longitude'].between(5.9, 6.3))
```

```
    ]
    for mmsi, group in df.groupby('mmsi'):
      if len(group) < 2:
        continue
      group = group.sort_values('date_time_utc')
      coords = [transformer.transform(lon, lat) for lon, lat in zip(group['longitude'], group['latitude'])]
      ship_line = LineString(coords)
      if ship_line.crosses(bridge_line):
        intersection = ship_line.intersection(bridge_line)
        if intersection.geom_type == 'Point':
          proj_dist = bridge_line.project(intersection)
          vec = np.array(coords[-1]) - np.array(coords[0])
          direction = np.dot(vec, bridge_vector)
          if direction < 0 and 0 <= proj_dist <= bridge_length:
            west_proj_dists.append(proj_dist)
  except Exception as e:
    print(f"Error reading {file}: {e}")

# === Compact KDE Domain ===
z_min_new, z_max_new = 1009.45, 1968.9
z_min_old, z_max_old = min(west_proj_dists), max(west_proj_dists)

def rescale_z(z, old_min, old_max, new_min=z_min_new, new_max=z_max_new):
  return new_min + ((z - old_min) / (old_max - old_min)) * (new_max - new_min)

compact_dists = [rescale_z(z, z_min_old, z_max_old) for z in west_proj_dists]

# === Compute KDE on compacted data ===
compact_kde = gaussian_kde(compact_dists, bw_method=0.05)
f_z_raw = lambda z: compact_kde(z) if z_min_new <= z <= z_max_new else 0

# === Normalize KDE ===
area, _ = integrate.quad(f_z_raw, z_min_new, z_max_new)
f_z = lambda z: f_z_raw(z) / area if area > 0 else 0
```

```
# === Compute Pcat2 using compacted KDE ===
P_0 = compute_P0(d, a)
total_Pcat2 = 0
results = {}

for (loa_min, loa_max), num_ships in ship_classes.items():
  Q_i = num_ships / T
  integrand = lambda z: P_0 * P_c * Q_i * f_z(z) * T
  integral, _ = integrate.quad(integrand, z_min_new, z_max_new)
  results[f"{loa_min}-{loa_max}m"] = integral
  total_Pcat2 += integral

# === Output ===
print("Category 2 Collision Probabilities by Ship Class (Westbound, Compacted Domain):")
for cls, val in results.items():
  print(f"{cls}: {val:.8f}")
print(f"\nTOTAL P_cat2 (Westbound): {total_Pcat2:.8f}")
```

8.4 A.4 Category 3 Risk – pcat3_west.py

```
#Pcat3
import numpy as np
from scipy.stats import gaussian_kde
import scipy.integrate as integrate
import matplotlib.pyplot as plt
import sys

# === Import precomputed KDE data for west crossings ===
sys.path.append(r"C:\path\to\AIS\data ")
from Fz import west_crossings # Ensure this list exists in Fz.py

# === Compacted KDE Domain ===
z_min_new, z_max_new = 1009.45, 1968.9
```

```
z_min_old, z_max_old = min(west_crossings), max(west_crossings)

def rescale_z(z, old_min, old_max, new_min=z_min_new, new_max=z_max_new):
  return new_min + ((z - old_min) / (old_max - old_min)) * (new_max - new_min)

compact_crossings = [rescale_z(z, z_min_old, z_max_old) for z in west_crossings]

# === KDE on compacted data ===
compact_kde = gaussian_kde(compact_crossings, bw_method=0.05)
f_z_raw = lambda z: compact_kde(z) if z_min_new <= z <= z_max_new else 0

# === Normalize KDE ===
area, _ = integrate.quad(f_z_raw, z_min_new, z_max_new)
f_z_kde = lambda z: f_z_raw(z) / area if area > 0 else 0

# === Parameters for Category 3 ===
P_e = 1e-3  # Probability of failed evasive maneuver
P_c = 0.0001 # Causation factor
T = 1    # Evaluation period in years

# === Ship traffic counts by LOA class ===
ship_classes = {
  (100, 125): 5,
  (125, 150): 15,
  (150, 175): 5,
  (175, 200): 5,
  (200, 225): 5,
  (225, 250): 5,
  (250, 275): 5,
  (275, 300): 10,
  (300, 325): 15,
  (325, 350): 30
}
```

```
# === Compute Pcat3 using compacted KDE ===
P_cat3_total = 0
ship_class_probs = {}

for (loa_min, loa_max), count in ship_classes.items():
  Q_i = count / T
  integrand = lambda z: P_e * P_c * Q_i * f_z_kde(z) * T
  integral, _ = integrate.quad(integrand, z_min_new, z_max_new)
  ship_class_probs[f"{loa_min}-{loa_max} m"] = integral
  P_cat3_total += integral

# === Results Output ===
print("\n=== Category 3 Collision Probability (Westbound, Compacted KDE) ===")
for k, v in ship_class_probs.items():
  print(f"{k}: {v:.6e}")
print(f"TOTAL P_Cat.3 (West): {P_cat3_total:.6e}")
print(f"Assumed Pe: {P_e}, Pc: {P_c}, Time period: {T} year")

# === KDE Plot ===
z_vals = np.linspace(z_min_new, z_max_new, 500)
kde_vals = [f_z_kde(z) for z in z_vals]
plt.plot(z_vals, kde_vals, label="Compacted f(z) KDE (Westbound)", color="green")
plt.xlabel("Distance along bridge axis from Anda (m)")
plt.ylabel("Density")
plt.title("Compacted KDE of Westbound Ship Crossings (f(z))")
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.show()
```

8.5 A.5 Heading Change Detection – heading_change_check

```
import pandas as pd
import numpy as np
```

```
import glob
from pyproj import Transformer
import math

# === Configuration ===
folder_path = r"C: \path\to\AIS\data "
files = glob.glob(folder_path + "/*.csv")
transformer = Transformer.from_crs("epsg:4326", "epsg:3857", always_xy=True)

angle_threshold = 10 # degrees difference in COG
distance_changes = []

for f in files:
  try:
    df = pd.read_csv(f, delimiter="\t", encoding='utf-8', on_bad_lines='skip')
    df.columns = [col.lower().strip() for col in df.columns]
    if not all(c in df.columns for c in ['latitude', 'longitude', 'cog', 'mmsi', 'date_time_utc']):
      continue
    df = df.dropna(subset=['latitude', 'longitude', 'cog', 'mmsi'])
    df['date_time_utc'] = pd.to_datetime(df['date_time_utc'], errors='coerce')
    df = df.sort_values('date_time_utc')

    coords = list(zip(df['longitude'], df['latitude']))
    cogs = df['cog'].to_numpy()

    for i in range(1, len(cogs)):
      delta_cog = abs(cogs[i] - cogs[i - 1])
      if delta_cog > angle_threshold:
        x1, y1 = transformer.transform(*coords[i - 1])
        x2, y2 = transformer.transform(*coords[i])
        dist = np.hypot(x2 - x1, y2 - y1)
        distance_changes.append(dist)
  except Exception as e:
    print(f"Error in file {f}: {e}")
```

```
if distance_changes:
  avg_a = np.mean(distance_changes)
  print(f"Estimated average a (using COG): {avg_a:.2f} m")
else:
  print("No significant heading changes (COG-based) detected.")
```